Main Supply (24 Power Distribution)

Li-Ion Battery 24V (6S, 3C); 20000mAh

+24V

GND

Mushroom Cap Emergency Stop Button

Emerrgency Stop Circuit

5V

5V and GND are routed to both the motor encoder terminals and the motor driver ground. These connections are omitted from the sketch for clarity due to wiring complexity.

XL4015 Buck

5V

24V

GND

200W 20A Buck Converter
24V step Down to 12V

24V - INPUT

12V - OUTPUT

XY3606 Buck Converter

5V

GND

+12V

As the USB interface was not supported in the diagramming environment, the connection is shown using pins as an alternative.

BNO055 IMU

Raspberry Pi5 Model B

In the hardware system, USB peripherals support the camera module, ESP32, and LiDAR connectivity.

Serial Interfaces
1. I2C: ESP32 to BNO055
2. UART: RPI ---> ESP32, Camera, and LiDAR

In the hardware design, power is delivered via a USB-C port and regulated using the XY3606 buck converter.

Maintaining a common ground is mandatory for proper operation

Connected Via MicroUSB cable to RPI5

ESP32 connections in the diagram are illustrative. The definitive pin mapping is provided in the text file within the 5.3 Electrical Schematics folder.

IG32 Motor Encoded

Cytron MD10C Motor Driver

10A Fuse

IG32 Motor Encoded

Cytron MD10C Motor Driver

10A Fuse

IG32 Motor Encoded

Cytron MD10C Motor Driver

10A Fuse

IG32 Motor Encoded

Cytron MD10C Motor Driver

10A Fuse

NEMA23 19kgcm Stepper Motor

ENABLE pin unused—TB6600 is enabled by default (active-low)

DIR-

PUL-

DIR+

PUL+

The hardware utilizes the TB6600 motor driver, which includes its own internal voltage regulator. Therefore, the VCC and GND pins for IC power on this board are not used.

fritzing